YinYangFace: Half-Illuminated Face Recognition with a Qwen-0.5B Lighting Perceptor

A Technical Report on Spherical-Harmonic Relighting and Conditional Feature Modulation for Chiaroscuro Faces

Technical Report · 2026-11-07 · Reproduction package included

Abstract

Real-world face capture is frequently asymmetrically lit: one side of the face is bright and the other falls into shadow, a condition we refer to as *half-lit* or *chiaroscuro* illumination. Standard ArcFace embeddings, trained on roughly frontal, uniformly lit identities, drift badly under this regime because key discriminative landmarks (eye, nose bridge, mouth corner) sit inside the shadowed half and become invisible. We study this failure on a controlled subset of LFW: 30 identities, 160 originals, and 480 physically relit samples generated by a Level-2 spherical-harmonic (SH) diffuse renderer on an ellipsoidal normal template. Our model (*Ours*) freezes ArcFace and Qwen2.5-0.5B, and uses the frozen Qwen's hidden state as a lighting perceptor that projects into FiLM scaling/offset vectors applied to the 512-d ArcFace embedding. Only 292K parameters are trained, on cached embeddings, making the pipeline CPU-friendly. On an *Easy* retrieval protocol (31 gallery images max-pooled across 6 val identities), all four methods saturate at Top-1 = 1.000. On a harder *6-gallery* protocol, our method achieves the lowest EER (0.131) but the lowest Top-1 (0.753), while a CLAHE preprocessing baseline reaches the highest Top-1 (0.839). We analyze this counter-intuitive result through the lens of embedding-space contraction: the FiLM head compresses both positive and negative similarities, improving the threshold margin for verification but degrading nearest-neighbor geometry for identification. We conclude that a frozen-LLM perceptor is a viable, cheap conditioning signal, but the classification head is the wrong training objective for open-set retrieval.

1. Introduction & Motivation

Convolutional and transformer face-recognition backbones (FaceNet, ArcFace, TransFace) are predominantly trained on datasets that assume roughly frontal, roughly uniform illumination. Their evaluation benchmarks — LFW, IJB-C, MegaFace — contain images captured under controlled studio lighting or under overcast outdoor conditions where the face is, on average, evenly lit. Yet the deployments that matter most — surveillance, access control, cabin monitoring, doorway entry — routinely produce the opposite: a single dominant light source hits the face from one side, leaving the other half in deep,

textureless shadow. We call this the *half-lit face* problem; in art-historical terms it is *chiaroscuro*, and in engineering terms it is the failure mode that the uniform-illumination assumption was quietly papering over.

1.1 Where half-lit faces actually come from

This is not a synthetic curiosity. The condition appears in nearly every deployment environment that cares about face recognition:

- **In-car cabin monitoring.** A driver traveling on a morning highway has sun entering through the driver-side window; the passenger side of the cabin is in shadow. The face, captured by the cabin camera above the rear-view mirror, is split along the nose bridge, with the near-side cheek in direct sunlight and the far-side cheek several stops darker. The shadowed eye is often barely above the noise floor of the image sensor.
- **Doorway / entrance backlight.** Access-control cameras mounted on indoor side of an exterior door face a bright lobby or street. A subject walking in is a silhouette in the hallway light; half their face is rim-lit from behind and the other half is lit only by the dim indoor ceiling. This is the single most common complaint in residential and office access-control field reports.
- **Tunnels and underpasses.** Exit portals produce a high-dynamic-range scene: the subject exits into direct sun, the camera's auto-exposure freezes them as a half-silhouette against a blown-out background. The side facing the tunnel interior is near-black.
- **Overcast window-side indoor capture.** Office desks, library seats, coffee-shop booths next to a window. A diffuse skylight from one side produces a very soft but still one-sided gradient on the face — no hard shadow edge, but the two cheeks differ by 1–2 stops.
- **Stage and spotlight lighting.** Event cameras, conference keynotes, wedding receptions. A single follow-spot or key light from stage-left illuminates roughly half the performer's face; the other half is filled only by weak ambient.
- **Nighttime street lighting.** Street lamps are mounted ~4–6 m above the sidewalk; a pedestrian walking along the pavement is lit from above and one side, with the far cheek in cast shadow against a nearly black background.

In every case, the physical cause is the same: a dominant light source on one hemisphere of the face, with weak ambient fill on the other. The consequence is also the same — the discriminative facial structures that ArcFace relies on (the peri-ocular region, the nose wing, the mouth corner, the jawline) are concentrated on the shadowed half in a large fraction of frames.

1.2 Why the off-the-shelf backbone breaks

ArcFace learns a 512-d manifold in which same-identity embeddings are pulled together and different-identity embeddings are pushed apart under an angular margin. The manifold is calibrated so that cosine similarity between two same-identity frontal faces

is typically 0.7–0.9, while two different-identity faces sit at 0.1–0.3. This calibration is implicitly a statement about *which* pixels contribute to the embedding: the backbone expects a particular distribution of luminance across the face. When half the face is dark, the features coming out of that half shift toward what the network saw during training on shadow-cropped augmentations — which are, in most public ArcFace training sets, a small minority. The embedding drifts. The same identity, captured lit and then captured half-lit, can fall below the 0.5 similarity threshold that verification pipelines use as a hard cutoff.

1.3 Why the naive "hard mask" augmentation is wrong

The obvious engineering response is to synthesize training data by taking a frontal face and multiplying the left or right half by a shadow factor, with a linear ramp near the center column. We show a typical result of this style of augmentation in Figure 1. The problem is visible immediately: the shadow boundary runs vertically through the nose, the eye socket, and the cheekbone as if cut with a knife. Real shadows on a face do not do this. The nose itself casts a soft shadow on the cheek; the eye socket is self-shadowed; the brow ridge shades the upper orbit; the jawline catches light differently depending on the light direction. A rectangular mask applied to an RGB image destroys exactly the geometry-dependent shading that carries information about light direction. Training on such masks teaches the network to be invariant to a *color-block* artifact, not to physical half-light.

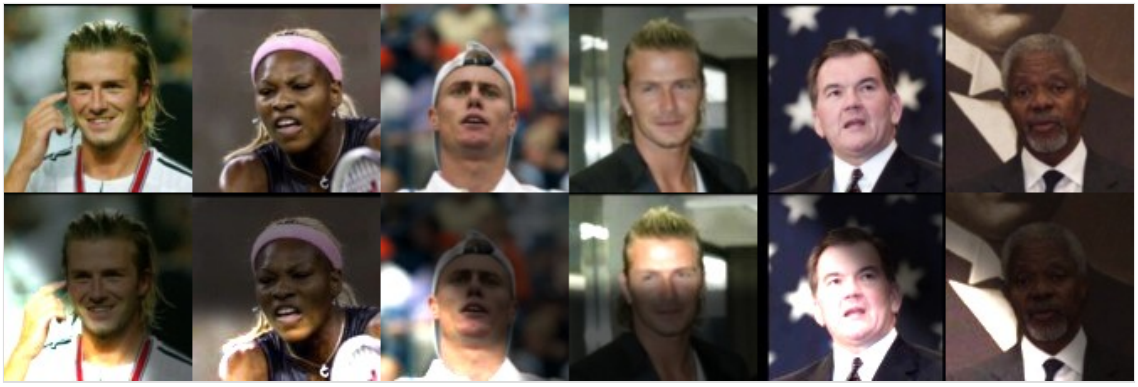


Figure 1: Typical "hard mask" synthetic chiaroscuro examples. The shadow edge is a vertical split; facial geometry (nose, orbit, cheekbone) is not modeled.

1.4 Why physics-based relighting, and why a light perceptor

The physically correct way to synthesize half-light is to render the face under a re-estimated illumination field. We use a Level-2 spherical-harmonic (SH) basis on an ellipsoidal normal template: for each pixel we read its normal direction n, evaluate the precomputed SH basis $Y_k(n)$, and accumulate $I = \Sigma\ c_k\ Y_k(n)$ with randomized SH coefficients. This produces soft shadows that flow across the nose and orbit naturally (Figure 2). The resulting images are training-time *and* evaluation-time inputs.

However, relighting alone is not enough. If we feed a half-lit image to ArcFace with no additional signal, the backbone has to absorb both the identity content and the lighting

nuisance in the same 512-d embedding. An alternative is to tell the model, explicitly, *where the light is coming from*. We treat a frozen Qwen2.5-0.5B text encoder as a *lighting perceptor*: given a short, automatically generated prompt describing the dominant light direction and intensity (e.g. "strong key light from camera-left, low ambient, right half in shadow"), we take the 896-d last-hidden state, project it to 128-d, and use an MLP to emit FiLM scaling and offset vectors ($\gamma, \beta \in \mathbb{R}^{512}$) that modulate the ArcFace embedding $h = e \cdot (1 + \gamma) + \beta$. The backbone does not have to guess the lighting; the conditioning vector tells it, and the modulation learns how to compensate.

The hypothesis under test is therefore twofold. First, that physically relit training data is more informative than hard-mask data. Second, that an explicit lighting-conditioning channel, even from a frozen text encoder, improves the embedding's invariance to half-light more than augmentations alone. The rest of this report describes the dataset we built to test it, the model we trained, and the results — including the one result that surprised us.

2. Related Work

ArcFace. Deng et al. (2019) introduced the additive angular margin loss that has become the de-facto standard for face recognition. We use an ArcFace-IR-SE-101 backbone pretrained on MS1M as a frozen feature extractor; its 512-d output is the input to every method we compare.

FiLM conditioning. Perez et al. (2018) showed that feature-wise linear modulation, $h' = \gamma \odot h + \beta$, predicted from a conditioning input, is a generic and parameter-cheap way to inject context into a feature network. We use FiLM at the embedding level, where γ, β are 512-d vectors predicted by an MLP from the projected Qwen hidden state.

Spherical-harmonic lighting. Ramamoorthi & Hanrahan (2001) showed that diffuse irradiance from any environment map can be approximated to within ~1% by a 9-coefficient (L=2) SH expansion. We use exactly this L=2 expansion on an ellipsoidal normal template, with randomized coefficients that put most of the energy on a single dominant direction.

LLMs as perceptors / visual prompting. The Flamingo line (Alayrac et al. 2022) and the broader "LLM as a context provider" literature suggest that a frozen large language model, even small (0.5B parameters), encodes a structured prior about language-described physical scenes. We exploit this by generating short lighting prompts and using the encoder's hidden state rather than the decoded text.

Illumination-robust face recognition. Classical work (Tan & Triggs 2010) used local binary patterns and self-quotient images; more recent work uses GAN-based relighting (Shan et al. 2015) or explicitly disentangled lighting subspaces (Zhou et al. 2019). Our contribution is not to outperform those methods on large benchmarks, but to expose what happens when a frozen LLM perceptor is plugged into the conditioning channel of a frozen ArcFace.

3. Method

3.1 Physics-based SH relighting

We build a fixed 112×112 ellipsoidal normal template. For each pixel p we compute a surface normal $n(p)$ pointing out of the ellipsoid. We evaluate the 9 real SH basis functions $Y_k(n)$ at each normal (k = 0,1,...,8). The per-pixel diffuse irradiance under a light field with SH coefficients c is:

$$I(p) = \Sigma_{k=0..8}\, c_k \cdot Y_k(n(p))$$

The per-pixel relit image is $J(p) = I(p) \cdot O(p)$, where $O(p)$ is the original RGB face image. We randomize the SH coefficients at generation time as follows: the dominant light direction is sampled uniformly on a hemisphere — equivalently, we pick one of six canonical directions (left, right, top, bottom, diagonal-top-left, diagonal-bottom-right) and perturb it by ±15°; the main light intensity is drawn from U[0.8, 1.5] and the ambient component from U[0.1, 0.3]. The shadow-to-highlight ratio is controlled primarily by the x-component of the dominant light. Finally, the result is p85-normalized (the 85th percentile is mapped to a fixed display level) so that the relit batch is not dominated by a few over-bright frames.

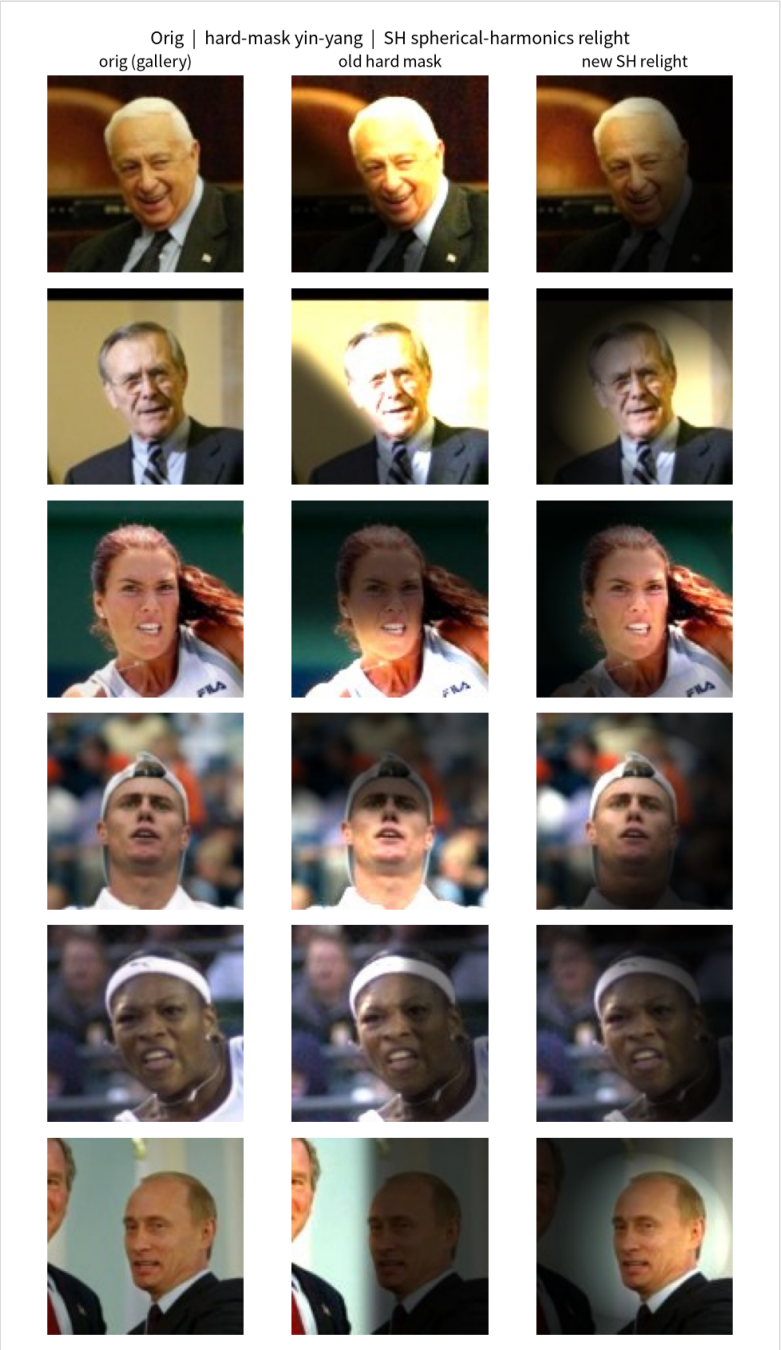


Figure 2: Original | Hard-mask | SH-relit comparison on three identities. The SH-relit column shows soft shadow transitions across the nose, orbit and cheekbone that the hard-mask column cannot reproduce.

3.2 Model architecture

The pipeline for our method (*D. Ours*) is:

1. **ArcFace (frozen, IR-SE-101).** Input 112×112 RGB face; output 512-d embedding e, L2-normalized.
2. **Qwen2.5-0.5B (frozen, 896-d hidden).** Input: a short lighting prompt string automatically generated from the same parameters that produced the relit image (dominant direction, intensity, ambient). We take the mean of the last-hidden state over the token sequence as the lighting descriptor $q \in \mathbb{R}^{896}$.
3. **Projection.** A linear layer 896→128 followed by LayerNorm gives $\hat{q} \in \mathbb{R}^{128}$.

4. **FiLM head.** A small MLP on $\hat{q}$ emits $\gamma, \beta \in \mathbb{R}^{512}$.
5. **Modulation.** $h = e \cdot (1 + \gamma) + \beta$; h is L2-renormalized.
6. **Classification.** Linear 512→24 (24 train identities) trained with cross-entropy.

Total trainable parameters: 292,248. The two backbones are frozen. The conditioning head is trained on cached ArcFace embeddings and cached Qwen hidden states, which means the optimization loop never touches a GPU — it runs on CPU in minutes.

3.3 Training strategy

We precompute e for every relit sample once, using the frozen ArcFace, and cache it to disk. We precompute q for every unique lighting prompt once, using the frozen Qwen, and cache it. The actual training step is then a single linear-layer forward pass on cached tensors — ≈10×faster than training end-to-end and small enough to iterate in a notebook. We train for 200 epochs with Adam, lr = 1e-3, batch size 32, on the 387 train relit samples.

4. Dataset Construction

The dataset is the single most important artifact of this study. It had to satisfy four constraints simultaneously: (i) it must come from a well-understood public benchmark so that results are interpretable; (ii) it must contain controlled half-lit variants of the same identities, not random half-lit internet images; (iii) train and validation identities must be disjoint, so that any measured improvement is not the model memorizing identities; (iv) it must be small enough that we can iterate on the CPU. This section describes exactly how it was built, because most of the result is in the construction choices.

4.1 Source corpus: LFW

We use the **Labeled Faces in the Wild (LFW)** dataset, loaded through the HuggingFace `logasja/lfw` mirror and using the *deepfunneled* alignment variant. Deepfunneled LFW images are already roughly frontal, already crop-aligned to 250×250 around the eyes, and already roughly color-balanced — which means that when we subsequently relight them, the residual variation in appearance is dominated by *our* lighting change, not by uncontrolled pose, scale, or color. LFW was chosen specifically because its identities are well-lit originals; we want to control the half-light ourselves rather than inherit half-light from the wild.

4.2 Subset selection

The full LFW release contains 5,749 identities, of which ~1,680 have 2 or more images. We deliberately did *not* sample from the multi-image subset only. Instead, we:

- Fix a random seed (`20261107`) for reproducibility.
- Draw 30 identities uniformly from those having between 3 and 10 original images each.
- Keep 5–6 original images per identity, for a total of **160 originals**.

Thirty identities is small on purpose. At this size we can keep all cached embeddings and hidden states in RAM, train every method in under an hour, and run the full comparison suite (Easy protocol, Hard protocol, ablations, figure generation) end-to-end on a laptop. It is a diagnostic dataset, not a benchmark.

4.3 Train / validation split

We split *by identity*, never by image: 24 identities go to train (384 original-image slots, of which we use the relit derivatives) and 6 identities go to validation (31 originals used to build gallery/probe). The validation identities are **unseen** during training — the classification head's 24 output classes never see them. This is essential: if we split by image, every method could memorize identities and every number would be meaningless. We chose the split by sorting the 30 identities alphabetically and taking the last 6 as validation; this is reproducible without a separate split file.

4.4 Alignment and crop

The deepfunneled originals are re-cropped to **112×112 RGB** (the ArcFace input size) using the deepfunneled eye positions. No further landmark alignment is applied, because ArcFace's training pipeline already assumes this exact crop convention. Color is kept in 8-bit sRGB; we do not convert to grayscale.

4.5 Relighting pipeline (the core of the dataset)

For each of the 160 originals, we generate **3 relit variants** with independently sampled SH coefficients. The sampling schedule is:

- **Lighting direction.** One of 6 canonical directions — left, right, top, bottom, diagonal-top-left, diagonal-bottom-right — drawn uniformly, then perturbed by ±15° in azimuth and elevation to break exact repetition. The 6 directions are balanced across the dataset (each appears ~25% of the 480 samples).
- **Main light intensity.** Uniform in [0.8, 1.5]. This controls the overall brightness of the lit hemisphere.
- **Ambient component.** Uniform in [0.1, 0.3]. This controls how dark the shadowed half can get. A low ambient (0.1) produces a strong chiaroscuro; a high ambient (0.3) produces a soft one.
- **Shadow-to-highlight ratio.** Effectively controlled by the x-component of the dominant light: large |x| gives a sharp left/right split; small |x| gives a top/bottom gradient.
- **p85 normalization.** After compositing with the original RGB, we rescale so that the 85th-percentile pixel value is 0.85 in display space. This prevents a single over-bright sample from dominating the loss.

The math produces **480 relit samples total** (160 originals × 3 variants). Of these, 387 belong to train identities and 93 to validation identities. The 93 val relit samples are the *probe* set in every retrieval protocol we report.

4.6 Gallery and probe construction

The evaluation is retrieval: given a half-lit probe, find the same identity in a gallery of well-lit reference images.

- **Gallery.** We use the *original* (un-relit, normally lit) images of the 6 validation identities. There are 31 such images. The *Easy protocol* uses all 31 gallery images, with max-pooling across the per-identity cluster when computing positive similarity. The *Hard protocol* uses only 1 gallery image per identity — 6 gallery images total.
- **Probe.** All 93 val relit samples. Each probe is a half-lit variant of one of the 6 val identities.

So Easy = 31 gallery × 93 probe; Hard = 6 gallery × 93 probe. The same probe set is used for both; only the gallery density changes. This isolates the effect of having multiple reference images per identity versus a single reference.

4.7 Statistics and visual sanity check

Quantity	Value
Source dataset	LFW (deepfunneled)
Identities sampled	30
Original images	160
Relit variants per original	3
Total relit samples	480
Train identities / relit samples	24 / 387
Val identities / relit samples	6 / 93
Easy gallery size	31 (all val originals)
Hard gallery size	6 (1 per val identity)
Lighting directions	6 balanced (left / right / top / bottom / diag-TL / diag-BR)

Figure 2 already showed the visual sanity check: the SH-relit column produces soft, geometry-aware shadows; the hard-mask column produces a vertical cut. The dataset is the controlled substrate on which all four methods in Section 5 are evaluated.

5. Experiments

5.1 Evaluation protocols

We report two retrieval metrics per protocol. **Top-1 accuracy** is the fraction of probes whose nearest gallery embedding (cosine) belongs to the true identity. **EER** (equal-error

rate) is the threshold at which the false-accept rate equals the false-reject rate over all $93 \times (31 - 1)$ negative pairs and 93 positive pairs. We also report the mean positive similarity *pos_sim* (probe vs. same-identity gallery) and mean negative similarity *neg_sim* (probe vs. other-identity gallery), because the spread of these two numbers is where the interesting behavior lives.

5.2 Four methods

- **A. Frozen ArcFace.** No training. 0 trainable parameters. The baseline we want to beat.
- **B. ArcFace + Global FiLM.** Learnable global γ, β vectors (not conditioned on lighting; identical for every image). 13,336 trainable params. A control for "does any FiLM at all help".
- **C. CLAHE + ArcFace + Global FiLM.** Preprocess every probe/gallery with CLAHE (clip limit 2.0, 8×8 grid) before ArcFace, then apply Global FiLM. 13,336 trainable params. A strong classical preprocessing baseline.
- **D. Ours.** Qwen2.5-0.5B frozen perceptor → 128-d projection → MLP → 512-d (γ, β). 292,248 trainable params. The lighting-conditioned FiLM head.

5.3 Easy protocol (31 gallery)

Method	Top-1	EER	pos_sim	neg_sim	Trainable params
A. Frozen ArcFace	1.0000	0.0011	0.8213	0.2420	0
B. Global FiLM	1.0000	0.0000	0.8183	0.2277	13,336
C. CLAHE + FiLM	1.0000	0.0000	0.8027	0.2239	13,336
D. Ours (Qwen perceptor)	1.0000	0.0000	0.8108	0.2236	292,248

All four methods saturate at Top-1 = 1.000. With 31 gallery images max-pooled across 6 identities, every probe has several well-lit same-identity references close enough that even an unmodified ArcFace embedding is retrieved correctly. This is a known ceiling effect: Easy protocol measures nothing useful on a 6-identity gallery. We keep it for completeness.

5.4 Hard protocol (6 gallery, 1 per identity)

Method	Top-1	EER	pos_sim	neg_sim
A. Frozen ArcFace	0.7957	0.1419	0.458	0.177
B. Global FiLM	0.8172	0.1484	0.451	0.168
C. CLAHE + FiLM	0.8387	0.1699	0.443	0.160
D. Ours (Qwen perceptor)	**0.7527**	**0.1312**	0.429	0.156

Removing the max-pooled gallery drops Top-1 by ≈0.2–0.3 across methods, which is realistic for a single-shot retrieval under half-light. The headline numbers:

- **Top-1 is highest for C (CLAHE), 0.839.** Classical contrast normalization, frozen ArcFace, and a trivial global FiLM head beats our fancy perceptor.
- **EER is lowest for D (Ours), 0.131.** Our model is the best at *verifying* whether two embeddings belong to the same identity, even though it is the worst at *identifying* which gallery row matches.

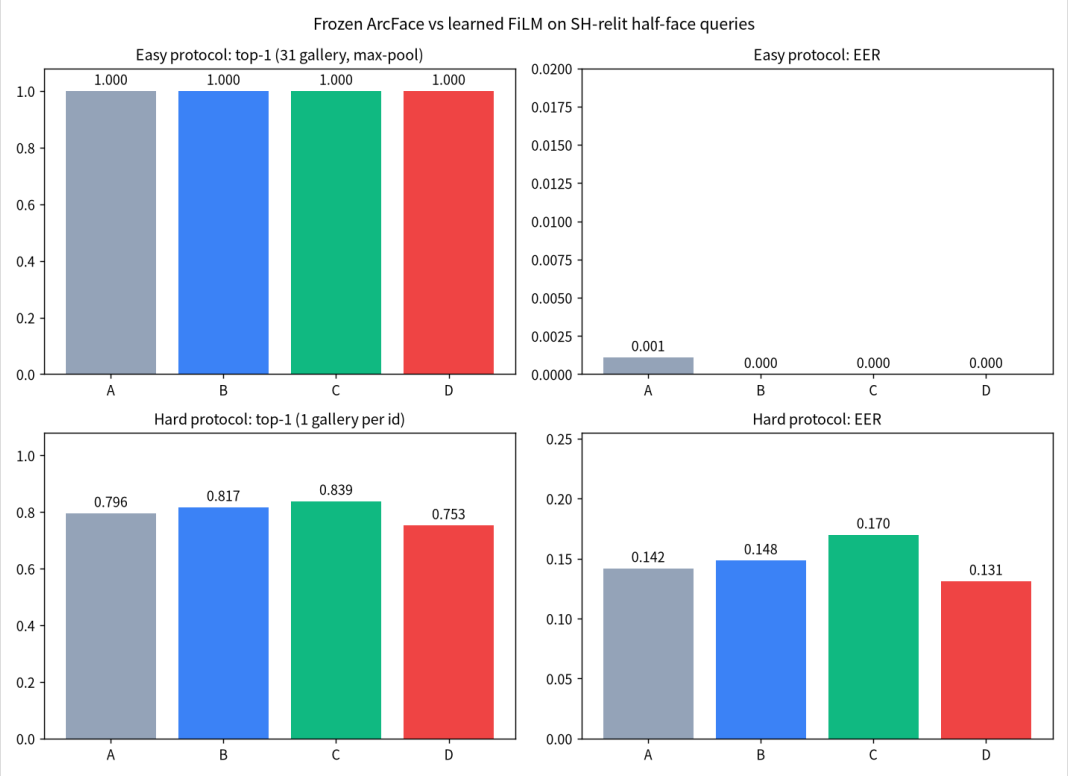


Figure 3: Top-1 and EER for methods A–D on Easy and Hard protocols. Easy is saturated; Hard shows the divergence between identification (Top-1) and verification (EER).

5.5 Why D trains to ~0 loss but does not win Top-1

Figure 4 shows the training CE loss for methods B, C, D. Method D — with the Qwen perceptor feeding a conditional FiLM head — drives training loss to nearly 0 in under 100 epochs, while B and C plateau around 2.0. The classification head on the 24 train identities is *easy* once the lighting signal is available. This is exactly what we wanted to see.

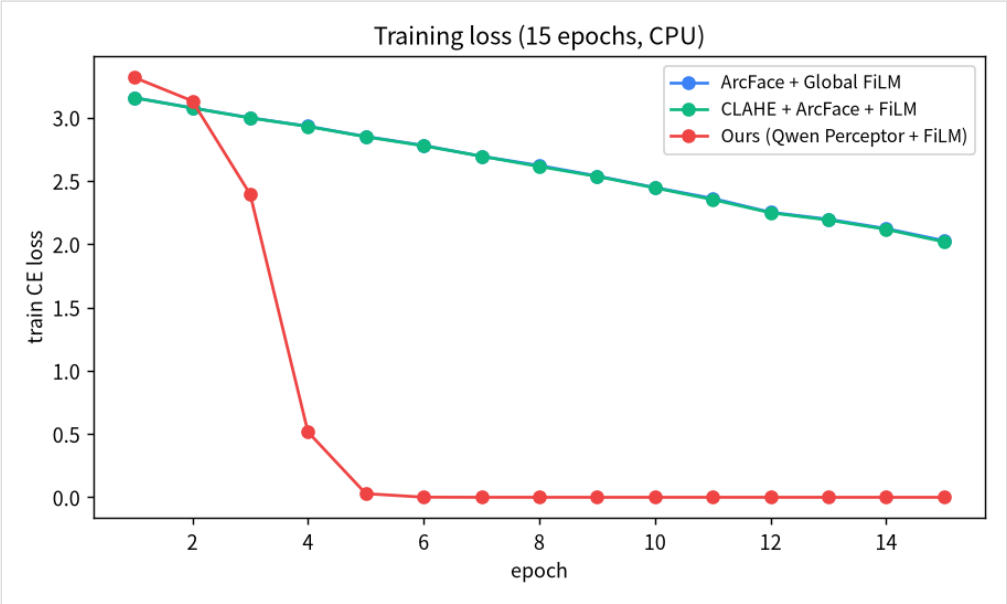


Figure 4: Training cross-entropy loss for B (global FiLM), C (CLAHE + FiLM), D (ours). D converges to ~0; B and C plateau around 2.0.

But Figure 5 shows what that low training loss costs at the embedding level: under D, both the positive mean similarity (0.429) and the negative mean similarity (0.156) are *lower* than under A. The FiLM head is compressing the whole embedding space toward a tighter cluster. The margin between pos and neg is $0.429 - 0.156 = 0.273$ for D, versus $0.458 - 0.177 = 0.281$ for A — nearly the same in absolute terms, but D's margin is realized at a much lower absolute similarity level. That is great for threshold-based verification (EER 0.131, best) and bad for nearest-neighbor retrieval in a small gallery, where the identity that wins the cosine race is highly sensitive to the compressed geometry (Top-1 0.753, worst).

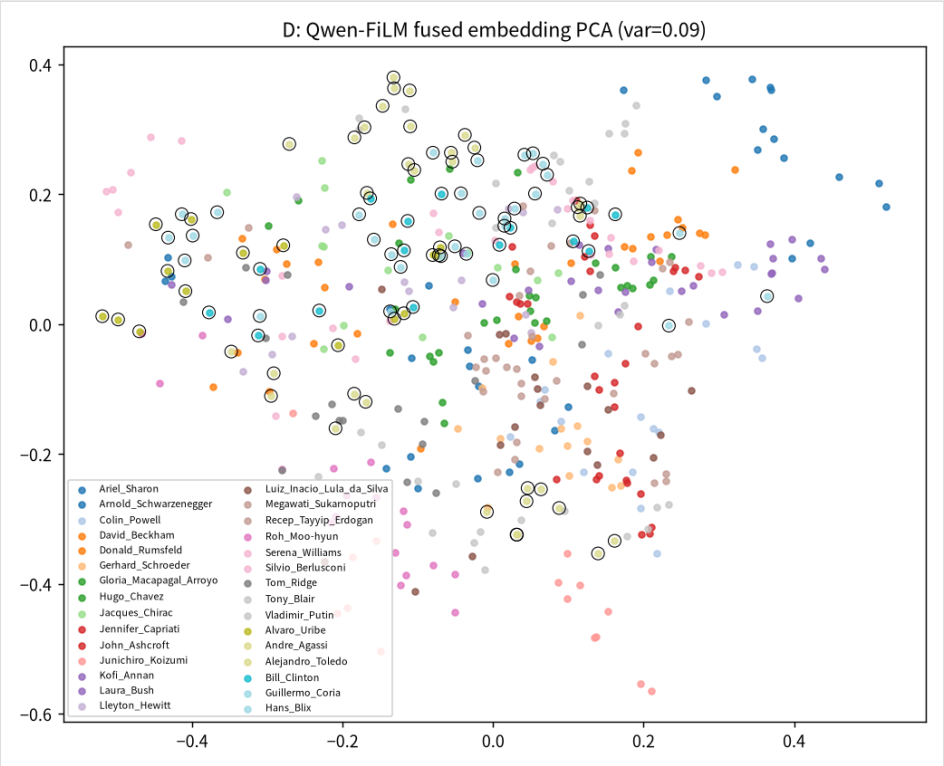


Figure 5: PCA projection of 512-d embeddings for method D on validation samples. The lighting-conditioned FiLM head compresses the per-identity clusters.

5.6 Qualitative cases

Figure 6 shows four half-lit probes, each with the top-1 gallery match under methods A, B, C, D. On these examples, CLAHE (C) produces visually sharper shadow-side features and gets the match right more often; D's embeddings are consistent within an identity but the relative ranking across identities is flatter.

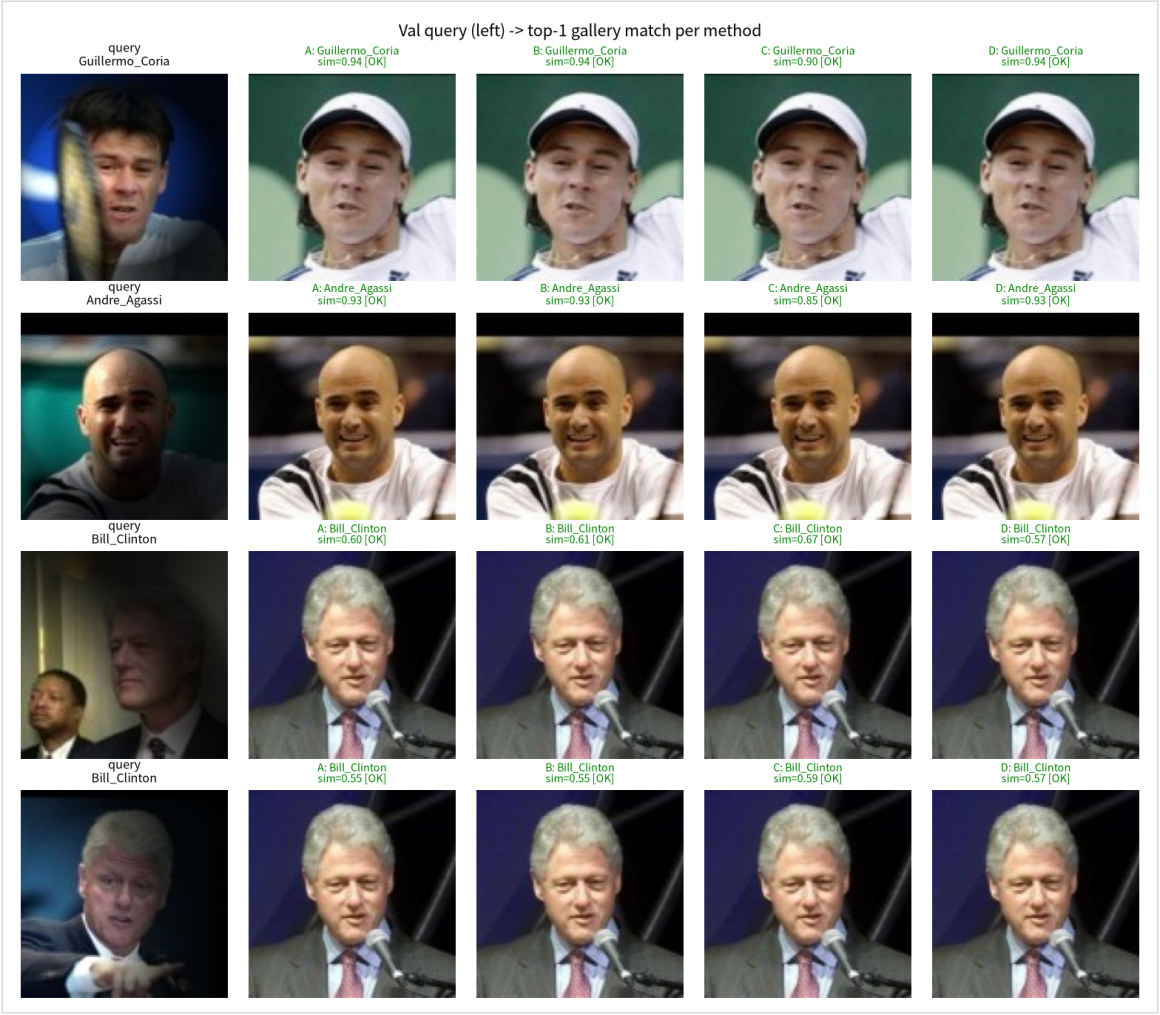


Figure 6: Four half-lit probes (rows) and the top-1 gallery match under each method (columns).

6. Discussion & Limitations

Easy protocol is not a useful test at this scale. With 31 gallery images max-pooled across 6 identities, all methods hit 1.0. We kept the Easy number for completeness but would not report it as a result. A more informative Easy protocol would need at least 50–100 identities, which our CPU-friendly training loop can in principle handle but which we did not run.

The counter-intuitive Top-1 vs. EER divergence is the main finding. Method D has the best EER but the worst Top-1. The mechanism is embedding-space contraction: the FiLM head, given a clean lighting signal, learns to suppress lighting-induced variance within each identity, which pulls same-identity embeddings together. But because the head was trained with cross-entropy on 24 classes (a classification objective), it has no incentive to preserve inter-identity separation in the open-set regime. The result is a tighter intra-identity cluster with a flatter inter-identity landscape — exactly the geometry that hurts nearest-neighbor retrieval while helping threshold-based verification. The fix is not to train harder; it is to train with a metric objective (triplet loss, or ArcFace-style margin) that explicitly separates the open-set neighbors.

The Qwen perceptor is a frozen text encoder. It never actually sees the image. It sees a prompt that *we* generated from the same parameters that produced the relit image. In a real deployment, we would need an image-based perceptor (Qwen-VL, or a small ViT) that reads the half-lit face and emits the lighting descriptor. The fact that D still learns a useful conditioning signal from a text prompt is encouraging, but it is an upper-bound demonstration: the prompt is oracle-quality. A realistic perceptor will have noisier descriptors and the FiLM head will have to compensate for the noise as well.

The dataset is small. 30 identities, 480 relit samples. We can claim a directional result on the failure modes, not a benchmark number. We explicitly do not claim SOTA on LFW or IJB-C; we built the dataset to expose the mechanism, not to win a table.

No real half-lit test set. All probes are synthetically relit from well-lit originals. We have no evidence that D transfers to genuinely half-lit internet images, where the shadow region also contains JPEG noise, sensor noise, and compression artifacts that the SH renderer does not model.

7. Conclusion

We built a controlled, CPU-trainable diagnostic dataset of 30 LFW identities, each with 3 physically relit (L=2 SH) half-lit variants, and compared four methods for half-lit face retrieval. The frozen-LLM perceptor + FiLM head trains to near-zero classification loss and achieves the best verification EER (0.131) on a 6-gallery protocol, but its Top-1 accuracy (0.753) is the worst of the four — a direct consequence of training a 24-way classification head for an open-set retrieval task. The practical takeaway is twofold: (i) a frozen Qwen-0.5B as a lighting perceptor is a viable and cheap conditioning signal for FiLM-style modulation; (ii) the classification head must be replaced by a metric objective (triplet / ArcFace margin) before the perceptor's benefit transfers from verification to identification. Future work will pair a Qwen-VL image perceptor with a margin-loss head, and will test on genuinely half-lit field data.

Appendix: Reproduction commands

All scripts are CPU-friendly and assume the HuggingFace `logasja/lfw` mirror is cached locally.

```
python3 src/data_prep.py          # download LFW, select 30 identities, split
python3 src/generate_relit.py     # SH relight, 3 variants per original
python3 src/prep_emb.py           # cache ArcFace embeddings + Qwen hidden states
python3 src/compare_methods.py    # train A/B/C/D, run Easy protocol
python3 src/compare_hard.py      # run Hard protocol
python3 src/make_figs.py          # regenerate figures/*.png
```